Giovanni **MINUTO**

Profile

I am an innovative and passionate Algorithms Developer with over 4 years of experience in the Machine Learning industry. My experience spans consulting, R&D, and applied research, with a strong focus on transforming advanced machine learning methods into reliable production systems.

Contact Details

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+39 373 5495 348

GitHub

My Portfolio for More Info

Google Scholar

Personal information

- Address: Turin, Italy
- Citizenship: Italian
- Year of birth: 1997
- Languages: English (C1), Italian (native)

Skills

- **Programming Languages:** Python, Bash, C++, MATLAB, LaTeX
- **Machine Learning & Data Science Libraries:** PyTorch, Scikit-Learn, TensorFlow, Keras, NumPy
- **Development Tools & Utilities:** Git, MLFlow, Docker, Multiprocessing, CI/CD pipelines, Slurm HPC manager
- **Cloud Platforms:** Amazon Web Services (AWS), Google Cloud Platform (GCP)
- **Methodologies & Practices:** Agile/Scrum

Work Experience

2025-2026

Machine Learning Engineer — *TMC, Italy*

- Development of data-driven, production-grade AI pipelines for enterprise clients to automate and accelerate existing business processes.
- Design of distributed machine learning architectures leveraging AWS cloud services.
- Applied machine learning for document intelligence, including OCR and structured information extraction.

2024–2025

Algorithms Developer — *Pasqal, Netherlands*

- Provided consultancy to LG Electronics, applying Physics-Informed Neural Networks (PINNs) to model and solve complex multi-physics problems.
- Built, trained, and evaluated machine learning models with a focus on expressivity, robustness, and trainability in industry-relevant use cases.
- Led a dedicated project vertical from system design to deployment.
- Developed and maintained multiple machine learning libraries for research and production use.

2023-2025

Teaching Assistant — *Polytechnic School of Genoa, Italy*

- Courses: *Fundamentals of Quantum Computing; Introduction to Quantum Information and Computation for Robotics.*

Education

2022–2025

PhD in Artificial Intelligence — *La Sapienza, Italy*

- Research focus: Bridging Quantum and Classical Machine Learning.

2020–2022

Master in Physics — *University of Genoa, Italy*

- Thesis: *Physics Simulations with Quantum Computers* (Grade: 110/110).

2016–2020

Bachelor of Science in Physics — *University of Genoa, Italy*

- Thesis: *Josephson junctions and superconducting materials* (Grade: 96/110).

Publications

2025

- *A Novel Approach to Reduce Derivative Costs in Variational Quantum Algorithms, (Journal of Physics A: Mathematical and Theoretical, IOP).*
- *Resource reduction for variational quantum algorithms by non-demolition measurements, (Preprint).*

2023

- *Quantum gradient evaluation through quantum non-demolition measurements, (European Physical Journal D, Springer).*

Open Source Libraries

Qadence:

- A simple interface to build digital-analog quantum programs on neutral atom devices.

Quantum Non-Demolition Measurement algorithm:

- Library for the QNDM algorithm, supporting gradient computation in quantum circuits.